

Predicting Overtakes in Formula One’s 2026 Regulation Era:

A Discrete-Time Hazard Model under Extreme Data Scarcity

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Abstract

Formula One’s 2026 regulations replaced the Drag Reduction System with an energy-based Overtake Mode, introduced active aerodynamics and moved to a 50/50 hybrid power unit. The mechanism of overtaking changed, so models trained on earlier seasons do not transfer, yet only fourteen races of in-era data exist. This paper addresses that cold-start problem. We reformulate overtake prediction as a *discrete-time hazard* over battle episodes with censoring, which removes the overlapping-label defect of a fixed-horizon classifier and recovers 706 observations a fixed-horizon formulation discards. We adopt an expanding-origin backtest that yields ten out-of-sample race results from fourteen rounds, and we transfer the pre-2026 era as a single frozen scalar rather than as pooled rows. Our extraction reproduces published season overtake totals to within 3.4%. The final model reaches PR-AUC 0.502 against a 6.58% base rate ($7.6\times$ lift), ROC-AUC 0.911 and Brier 0.0438. Of eight design hypotheses tested, six were overturned by measurement: a historical circuit passability prior did not transfer ($\rho = +0.09$), monotone constraints had no effect, hyperparameter tuning cost 1.2%, and removing seven features gained 5.1%. Two held: an old-era model supplied as one feature gained 4.2% while improving nine of ten rounds, and circuit *physics* transferred ($\rho = +0.97$) where circuit *outcome* did not. A recurrent model over episode sequences did not beat gradient boosting. We report all of it.

1 Introduction

Given two Formula One cars fighting on track, what is the probability that the car behind completes a pass? The question is well posed and the data to answer it is, in most seasons, plentiful. The 2026 season is not most seasons.

Three properties define the problem and drove every decision in this work.

The mechanism changed. The 2026 regulations [7] removed DRS and replaced it with an energy-based Overtake Mode, introduced active aerodynamics with distinct straight and corner modes, and split the power unit evenly between combustion and electrical power. Cars are 30 kg lighter with a 200 mm shorter wheelbase. A model trained on 2022–2025 does not transfer unmodified, and a feature set built around DRS zones describes a mechanism that no longer exists.

There is almost no in-era data. At the time of writing, fourteen races had been completed, yielding 470 passes across 6 669 usable observations. For comparison, the same pipeline applied to 2022–2025 produces roughly 49 000 rows.

The era is ongoing. A race is added every fortnight. The interesting question is therefore not “what is the best model on a fixed dataset” but “how should a model be built, evaluated and updated while the data keeps arriving”. This motivates a walk-forward evaluation protocol and a retraining loop rather than a single split.

Contributions.

- (i) A discrete-time hazard formulation of overtake prediction with proper censoring, which fixes three defects of the fixed-horizon framing (§4).
- (ii) An evaluation protocol that extracts ten out-of-sample results from fourteen races and quantifies its own noise floor (§5).
- (iii) A transfer mechanism that compresses four prior seasons into one frozen scalar, with measured gains and an ablation (§6).
- (iv) A tested distinction between circuit properties that survive a regulation change and those that do not (§6.3).
- (v) Six overturned hypotheses reported alongside two confirmed ones (§13).

Materials

Report	this document (era2026/paper/report.pdf)
Code	https://github.com/AminN77/formula1-overtake-prediction installable package <code>era2026/</code> , 88 tests
Interactive results	era2026/ui/index.html (opens directly, no server)
Design record	era2026/DESIGN.md , era2026/PLAN.md
Data reconnaissance	era2026/recon_report.md
Figure generation	era2026/paper/figures.py , numbers.json
Prior project	legacy/ at tag <code>legacy-v6</code>

Every figure and every number quoted in this report is produced by `paper/figures.py` from a run of the pipeline, and regenerates with the model; none are transcribed by hand.

2 Data

2.1 Source and reconnaissance

All data comes from FastF1 [12], which wraps Formula One’s live timing service. Only lap records, weather and race control messages are read; car telemetry is never required, because the speed-trap measurements used here are lap-level fields.

A reconnaissance pass established what 2026 actually exposes before any modelling decision was taken. All fourteen completed rounds load (16 255 lap records, 22 drivers, 11 teams) and the lap schema is identical to 2024. Critically, DRS is confirmed *dead rather than deprecated*: the telemetry channel still exists but all 35 553 samples read zero, against a live distribution in a 2024 reference race. No telemetry channel exists for Overtake Mode, active aerodynamics or energy deployment.

2.2 Defining and validating an overtake

An overtake is detected as a *pairwise order flip*: driver *A* is behind driver *B* at the end of lap *L* and ahead at the end of lap *L* + 1. This is the only definition available from timing data that references neither DRS, aero nor energy, which is precisely why it survives the regulation change.

Raw flips overcount, because pit stops and retirements reorder the field without an on-track pass. Pit-involved flips are excluded; lap one, safety-car periods and inaccurate timing are retained, as they contain real passes.

Table 1 is the checkpoint on which everything downstream depends.

Table 1: Extracted overtake counts against published figures, first four rounds of 2026. The same detector applied to 2025 gives 77 against a published 84, so the method lands within roughly 10% in both eras.

Circuit	Published	Extracted	Difference
Australian	39	37	−2
Chinese	71	73	+2
Japanese	43	42	−1
Miami	50	58	+8
Total	203	210	+3.4%

2.3 Reconstructing Overtake Mode

Overtake Mode surfaces only as lap-stamped race control messages, and reconstructing its state is less direct than it appears. Race control announces an explicit disable only at the standing start; later neutralisations disable the mode *silently*, and the re-enable is announced inconsistently. Canada takes virtual safety cars on laps 31, 46 and 53 and logs a re-enable after each; Australia takes them on laps 12, 18 and 34 and logs none. Replaying messages alone leaves Australia switched off for its final 46 laps.

The rule that survives all fourteen rounds uses track status as the primary signal with messages as explicit overrides. Overtake Mode is available on 727 of 855 race laps (85.0%).

3 Exploratory analysis

3.1 Overtaking rose sharply and unevenly

On the thirteen circuits raced in both seasons, passes rose 72% against 2025. The change is not a uniform multiplier: Monza more than tripled ($3.10\times$) while Barcelona fell to $0.56\times$. This single observation drove the design decision in §6.3.

3.2 Passes happen from very close range

Figure 1 shows the gap distribution on the lap preceding a pass. The median is 0.43s and the 99th percentile 2.36s. This fixed the candidate threshold empirically: a 3.0s window captures 98.9% of passes, and widening to 4.0s buys a further 0.3% at the cost of 1 161 additional rows.

3.3 The target is rare and varies tenfold by circuit

The overall base rate is 7.05%, but per race it ranges from 1.1% at Monaco (7 passes from 109 episodes) to 11.8% at Monza (69 from 217). Any evaluation reporting a single race’s score measures the circuit as much as the model.

4 Problem formulation

The natural formulation—“given this battle, will a pass occur within three laps?”—has three defects.

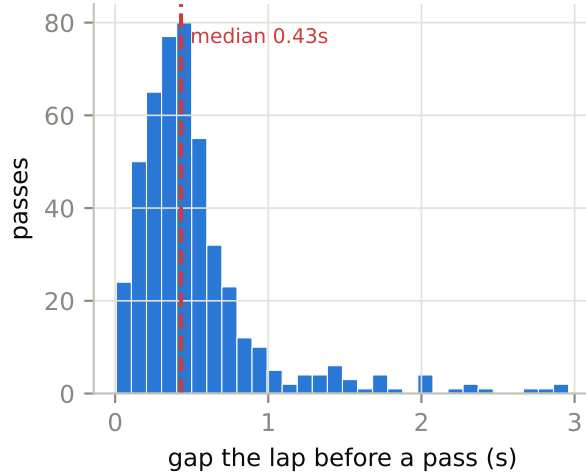


Figure 1: Gap on the lap before a completed pass. Overtaking is a close-range phenomenon; the long tail is what makes the candidate threshold a real choice rather than an obvious one.

- (a) **Overlapping labels.** A pair battling on laps 12–14 with a pass on 15 yields three rows, all labelled positive, with near-identical features. The model sees three examples and receives roughly one example of information.
- (b) **Discarded information.** If the defender pits, the pass *cannot* occur. Deleting such rows discards the genuine evidence that the pair fought for two laps without a pass.
- (c) **A frozen horizon.** Three horizons require three label columns.

We instead model the per-lap hazard, following standard discrete-time survival practice [15, 16]:

$$h(t) = P(\text{pass on lap } t+1 \mid \text{pair still battling at lap } t). \quad (1)$$

Any horizon follows by composition,

$$P(\text{pass within } k \text{ laps}) = 1 - \prod_{j=1}^k (1 - h_j), \quad (2)$$

so the horizon becomes a query parameter rather than a training decision.

Each row takes one of three outcomes. An **event** is a filtered on-track pass between L and $L + 1$. **Survived** means no pass, and deliberately includes the attacker dropping out of range, which is a genuine failure rather than a missing observation; censoring it would bias the hazard upward by discarding the clearest negatives available. **Censored** means the opportunity was removed rather than declined—either car pitted, either car retired, or the race ended. Censored rows carry no label but every earlier lap of the episode is retained.

The resulting dataset comprises 860 episodes, 7 491 rows of which 6 669 are trainable, 470 events at a 7.05% positive rate, and a mean episode length of 8.71 laps. Censoring retains 706 pit-affected rows that a fixed-horizon formulation deletes outright.

5 Evaluation protocol

The protocol was fixed *before* any model was trained.

Expanding-origin backtest. For each round k from 5 to 14, we train on rounds $1 \dots k-1$ and predict round k [9]. Every round becomes an out-of-sample test exactly once, so fourteen

rounds yield *ten* test results rather than one. A purpose-built test instruments the harness with a model that records which rounds it trained on, verifying that round k never appears in its own training set.

Frozen hyperparameters. A 40-trial random search [2] ran on rounds 1–4 only, and the winner was written to a committed file. A configuration chosen by inspecting round k makes round k no longer out of sample.

Metrics. PR-AUC is primary: the target is rare, and ROC-AUC flatters classifiers under class imbalance [5]. We also report ROC-AUC, the Brier score [3] and expected calibration error, because the model’s output is displayed as a probability and ranking quality alone does not establish that a stated 20% means twenty percent.

The noise floor. Per-round PR-AUC has a standard deviation of **0.228**. This figure governs how every subsequent result must be read: a difference of 0.02 between two models is not a difference.

6 Features and transfer

6.1 Levels from 2026, shapes from the old era

The overtake *rate* is directly measurable from 2026 alone and belongs in the intercept. What earlier seasons can supply is *conditional* structure: given a battle, what makes a pass likely. Pooling the two eras naively violates this, because a single shared intercept across two base rates invites the model to explain the difference using stale car performance.

Features are therefore tagged by transfer tier. **Tier 1** (regulation-invariant) covers race state, positions, Overtake Mode state, validated circuit character, and cross-race form computed from prior rounds only. **Tier 2** (same meaning, shifted relationship) covers gap and gap dynamics, pace and speed-trap *deltas*, tyre and stint state. **Tier 3** is excluded by construction and a unit test enforces its absence: all DRS features, all *absolute* car speeds and lap times, and team identity. Team identity is dropped because 2026 fields eleven teams of which two have no history and the remainder carry a performance level that is no longer true.

6.2 The transfer mechanism

The pre-2026 era enters as *one frozen scalar per row*: the output of a gradient-boosted model [10] fitted once on 48 932 legacy rows across 2022–2025, reading the 43 features both eras share and targeting a next-lap pass to match Equation 1. It never sees a 2026 row, so it cannot leak into any fold and requires no refitting. Four seasons therefore reach the model as a single column, and disabling it makes the ablation exact.

6.3 Circuit physics transfers; circuit outcome does not

An earlier design collapsed the circuit into a *historical passability prior*: a per-circuit overtake rate fitted on 2022–2025, on the argument that levels shift but the ranking is geometry. Measured against the thirteen comparable circuits, the Spearman rank correlation was $\rho = +0.09$ ($p = 0.76$), and per-circuit ratios spanned 0.56–3.10. The hypothesis was rejected.

The mechanism explains the failure and supplies the replacement. Under DRS, passability tracked zone length and slipstream; Overtake Mode is an *energy* aid, so what matters is full-throttle time. We therefore transfer circuit *physics* rather than circuit *outcome*—a historical overtake rate encodes what happened under a ruleset that no longer exists, whereas a speed

trap encodes the shape of a track the cars still drive. Crucially, the claim was tested before being used (Table 2).

Table 2: Rank correlation of per-circuit quantities between 2022–2025 and 2026, across the thirteen circuits raced in both eras. Physics transfers; top speed—the one property the new power unit and active aerodynamics actually changed—does not, and is excluded.

Quantity	Spearman ρ	p	Decision
Mid-sector speed trap	+0.974	< 0.0001	retained
Straight dominance (top / mid-sector)	+0.967	< 0.0001	retained
Sector-1 speed trap	+0.966	< 0.0001	retained
Race distance	+0.999	< 0.0001	retained
Top speed	+0.461	0.11	excluded
<i>Historical overtake rate</i>	+0.09	0.76	rejected

7 Models

We compare a non-ML baseline, two linear models, two tree ensembles, a kernel method, a feed-forward network and a recurrent model. Gradient boosting uses monotone constraints encoding known physics. The comparison is reported in full in §8; the recurrent model is treated separately in §12.

8 Results

Table 3: Pooled over ten backtest rounds (4 680 rows, 308 events, 6.58% base rate). Lift is PR-AUC relative to the base rate. Against a per-round standard deviation of 0.228, the tree ensembles are clearly ahead of everything else while the choice *within* that family is not resolvable.

Model	PR-AUC	Lift	ROC-AUC	Brier
Base rate	0.057	0.9×	0.440	0.0617
Gap alone	0.323	4.9×	0.851	0.0514
Logistic (5 features)	0.370	5.6×	0.874	0.0495
SVM (RBF)	0.418	6.3×	0.863	0.0531
Multilayer perceptron	0.444	6.7×	0.878	0.0488
GRU (pretrained)	0.453	6.9×	0.876	0.1018
Logistic (all features)	0.462	7.0×	0.884	0.0460
Random forest	0.493	7.5×	0.906	0.0457
LightGBM	0.515	7.8×	0.909	0.0448
LightGBM + isotonic (shipped)	0.502	7.6×	0.911	0.0438

Table 3 and Figure 2a give the model comparison. The shipped model applies isotonic calibration [17], surrendering a little PR-AUC for the best ROC-AUC, the best Brier score and a threefold reduction in calibration error—the correct trade when the output is displayed as a probability. Platt scaling [13] was tested and proved *worse than no calibration* on this data (ECE = 0.029 against 0.014 uncalibrated), and is not used.

8.1 The transfer layer

Supplying the pre-2026 era as one frozen scalar improves PR-AUC from 0.470 to 0.490 (+4.22%) and improves **nine of ten rounds**. Given the noise floor, the consistency carries more evidential

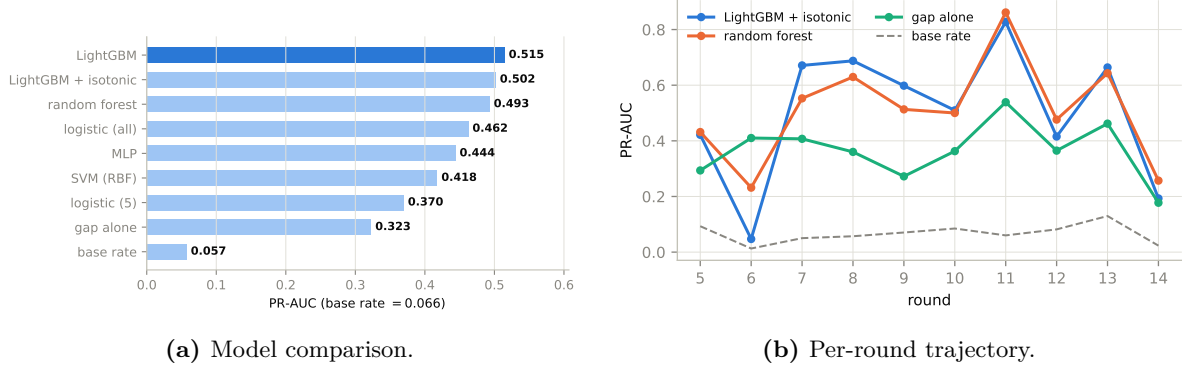


Figure 2: Left: pooled PR-AUC by model. Right: PR-AUC for each backtest round. The per-round variance ($\sigma = 0.228$) dwarfs the differences between the leading models, which is why no single race can adjudicate between them.

weight than the magnitude: nine of ten would arise by chance approximately 1% of the time.

The result of interest is not the magnitude. The old-era model is *weaker alone than a single in-era feature*—ROC-AUC 0.768 against 0.862 for raw gap—yet it still improves the full model. It is therefore not re-supplying gap; it carries structure the in-era model cannot recover from fourteen races.

8.2 Calibration and uncertainty

Figure 3a shows the reliability curve of the shipped model. Isotonic calibration reduces expected calibration error from 0.0142 to 0.0049.

Epistemic uncertainty is estimated by a bootstrap [6] resampled at *race* level, since rows within a race are not independent. Raw interval width is confounded by each test round’s own hazard level, so width is normalised by the predicted hazard. Interval width then narrows significantly as the era accumulates data (Spearman $\rho = -0.649$, $p = 0.043$; Figure 3b). This is deliberately *not* an accuracy claim: see §11.

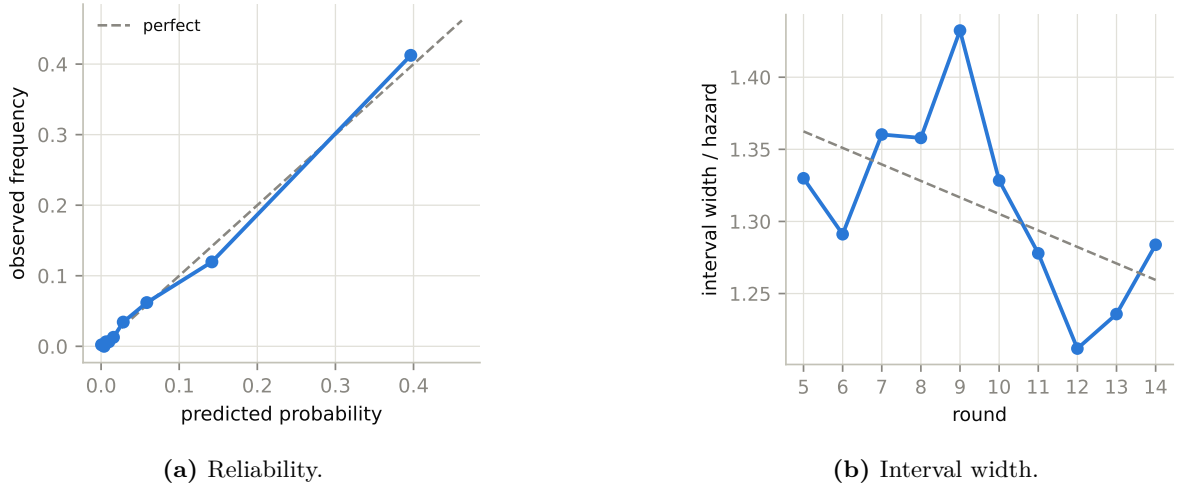


Figure 3: Left: predicted against observed frequency; points on the diagonal are honest probabilities. Right: bootstrap interval width relative to the predicted hazard, which narrows as rounds accumulate.

9 Explainability

Permutation importance [1] measured on held-out rounds is cross-checked against TreeSHAP [11]. Both rank `gap_ahead` first and the *transfer scalar* `old_era_score`, which is independent confirmation of the ablation in §6. All SHAP directions are physically correct: a larger gap lowers the hazard, closing raises it, a faster attacking team raises it.

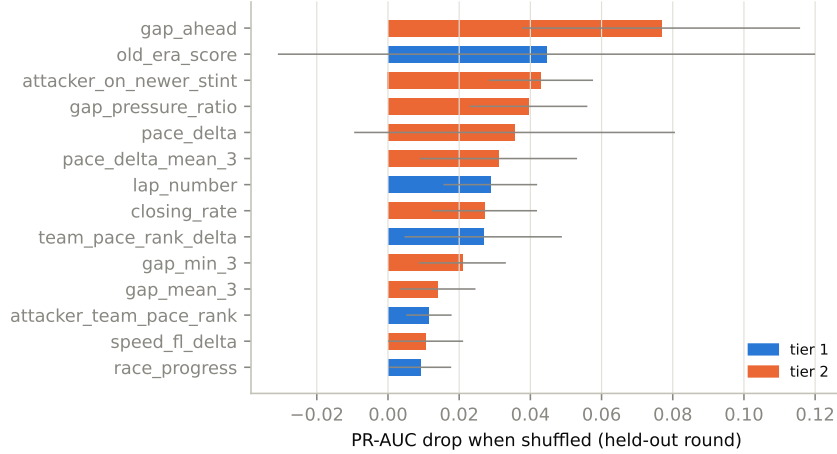


Figure 4: Permutation importance on a held-out round, with standard deviation across five shuffles. The transfer scalar `old_era_score` ranks second of 57 features.

The feature set was over-engineered. Of 57 features, 23 have zero or negative permutation importance. Removing the seven features that are *constant within a race*—weather, lap count, career counters—raised PR-AUC from 0.489 to 0.515 (+5.1%). The model had been using them as circuit fingerprints: memorisation of which race it was looking at, which cannot generalise to an unseen circuit. This was the largest single improvement in the project after the transfer layer, and it came from *removing* information. The validated `circuit_*` features are retained despite also being race-constant, because they encode character shown to transfer rather than an incidental identifier.

10 Error analysis

Sliced by circuit, the failures are concentrated and point one way (Figure 5). The model predicts 6.8% at Monaco where the truth is 2.0%, and 6.9% at Monza where the truth is 12.7%. **The two extremes of circuit passability are precisely where the model breaks, in opposite directions.**

Adding validated circuit physics (§6.3) raised Monaco’s PR-AUC from 0.096 to 0.179 but barely moved the bias. Speed traps capture how *fast* a circuit is, not how *passable*. Monaco is too narrow to pass on at any speed, and track width, braking-zone count and corner-exit geometry are not derivable from timing data at all. Closing this gap requires an external geometry source and is the clearest single item of future work.

By gap band, calibration is otherwise excellent, with bias below 0.007 in every band.

11 Convergence and overfitting

Training on a growing prefix of rounds while always scoring the same held-out race isolates training size from which race is predicted (Figure 6). Two conclusions follow.

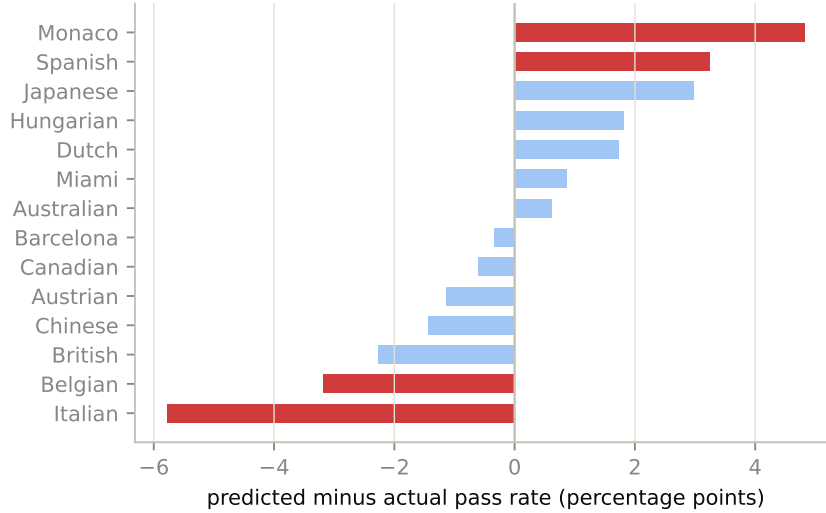


Figure 5: Predicted minus actual pass rate by circuit. Highlighted bars exceed three percentage points. Monaco and Monza—the extremes of passability—fail in opposite directions, which is the measurable cost of excluding circuit identity.

First, the model **overfits heavily**: training PR-AUC sits near 0.96 at every training size against held-out values near 0.25, a generalisation gap above 0.65 throughout. Second, and more importantly, **additional data has not improved accuracy**. Held-out performance is flat from two training rounds (0.279) to thirteen (0.252), consistent with the per-round trajectory in Figure 2b, which shows no trend. The extra data bought narrower *uncertainty* (Figure 3b) and nothing else. The feature reduction reported above is the natural response; stronger regularisation is the obvious next step.

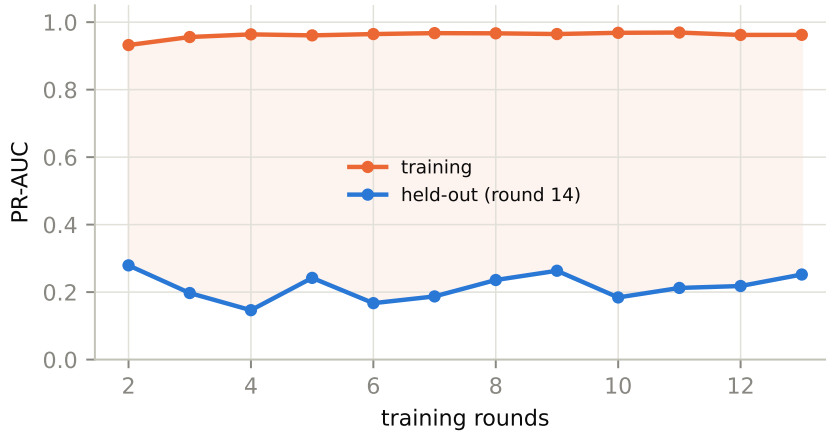


Figure 6: Learning curve. The shaded region is the generalisation gap. Training performance is near-perfect at every size while held-out performance is flat, indicating a model limited by the informativeness of the data rather than by its quantity.

12 The sequence model

Every other model treats a battle-lap as an independent row carrying hand-computed history. A GRU [4] over the episode learns its own summary instead, with identical features, target and protocol; the network is unidirectional and the hazard head reads the hidden state at lap t , so

no future information leaks in.

It does not win (Table 3), which is the expected outcome on heterogeneous tabular data at this sample size [8, 14]. Two details matter more than the headline. On *ranking* the gap to LightGBM is within the noise floor; where the GRU is clearly worse is *calibration*, with a Brier score more than twice as poor, which disqualifies it for a probability display regardless of ranking. Second, pretraining the encoder on four prior seasons gained 0.3%, against 4.22% for the same era supplied as a frozen scalar feature. **At this scale the old era transfers as a feature, not as a representation**—the opposite of what the design predicted.

13 Hypotheses overturned

Eight design hypotheses were stated in advance and then tested. Six were overturned (Table 4), and several of the failures produced the largest improvements in the final model.

Table 4: Design hypotheses against measured outcomes.

Hypothesis	Outcome
A historical circuit passability prior transfers	Rejected. $\rho = +0.09$, $p = 0.76$
Monotone constraints help at small n	No effect. 0.4662 vs 0.4641
Cross-race form is the largest remaining gap	Marginal. +0.8%; in-race pace already subsumes it
Hyperparameter tuning improves the model	Rejected. -1.2%; the search selected noise
Pretraining is where a sequence model wins	Rejected. +0.3%
More features are better	Rejected. Removing seven gained +5.1%
An old-era model as one feature is the best transfer	Confirmed. +4.22%, better in 9/10 rounds
Circuit <i>physics</i> transfers where <i>outcome</i> did not	Confirmed. $\rho = +0.97$

The frozen tuned parameters remain the default despite being 1.2% worse than a hand-chosen prior, because adopting the prior *after* observing it win on the test rounds is exactly the post-hoc selection the protocol exists to prevent.

14 Reproducibility and operations

The project is an installable Python package with 88 tests. A retraining loop is implemented and runnable but deliberately dormant: the incumbent scores each new race *before* anything is retrained, so evaluation and drift detection are a single computation; the promotion gate compares *configurations* walked forward over five rounds rather than fitted objects over rounds the challenger has already seen; and the drift monitor reports only what its sample size supports, holding race-level features silent until a second season exists.

15 Limitations

1. **Fourteen races.** 470 events is a small sample and every conclusion should be read against $\sigma = 0.228$.
2. **Circuit passability remains unrepresented.** Physics was validated and helped, but width, braking zones and corner-exit geometry are not obtainable from timing data.
3. **Overtake Mode is observable but not identifiable.** Its availability correlates with track neutralisation at -0.954 : the aid is withdrawn essentially only when passing is

forbidden anyway, leaving 49 of 6 669 rows as green-track-with-mode-off. The feature is retained but credited with nothing.

4. **Per-car energy deployment is invisible.** Whether a driver had energy banked and chose to spend it is not exposed in any channel.
5. **No cross-era validation of the transfer claim.** The 4.22% is measured within 2026 only.

16 Conclusion

Overtake prediction in a new regulation era is a cold-start problem, and this work treats it as one. Reformulating the target as a discrete-time hazard with censoring recovers information a fixed-horizon classifier discards; an expanding-origin protocol extracts ten out-of-sample results from fourteen races and quantifies its own noise floor; and compressing four prior seasons into a single frozen scalar transfers useful structure while leaving the new era’s base rate untouched.

The most useful outcome may be the negative results. Six of eight stated hypotheses were overturned, and the two largest improvements in the final model came from *rejecting* a designed component and from *removing* features. The model is limited by the informativeness of fourteen races rather than by their quantity, and its failures are concentrated at precisely the two circuits whose character it was forbidden to memorise.

Declaration of LLM use

A large language model assistant was used for implementation, debugging and drafting, in accordance with the course’s fair-use guidance. It was used to write and refactor code, to draft this report from the project’s own measured results, and to propose analyses. Every methodological claim herein is supported by a measurement recorded in the project’s version history rather than by assertion, and the project’s rejected hypotheses are reported alongside its confirmed ones. The author is responsible for, and able to justify, every choice, parameter, result and interpretation in this report.

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